RESEARCH-ARTICLE

Natural Language-Driven AI Programming: Exploring a Gamified Learning Method with Broad Application Potential

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Abstract

Rapidly developing AI programming tools provide a new possible way for gamified learning. Users can input prompts to AI programming tools to edit games. Owing to the current limitations of AI abilities, it is unfeasible to edit games with high complexity.However, generative AI tools empower individual to engage in programming by lowering technical barriers, which facilitates a broad range of opportunities for individuals lacking programming expertise to create and modify educational video games. Such applications can similarly foster education equity.[1]In this paper, we present a new approach through the Rosebud AI platform employing AI programming to design an interactive scenario-based game to simulate the application scenarios of medical laws. In comparing the traditional teaching (n=100) to gamified teaching (n=101), the gamified teaching proved more effective in stimulating students' activity ($t=4.87$, $p<0.01$, $d=0.69$), knowledge comprehension ($t=5.34$, $p<0.01$, $d=0.75$), and learning motivation ($t=4.96$, $p<0.01$, $d=0.70$).Experiments show that the use of AI programming tools in education of law has great application value, which can effectively improve students' cognitive and application ability.

CCS Concepts

• **Applied computing** → Education; Interactive learning environments.

Keywords

Artificial Intelligence, Gamified Learning, Medical Law

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1 INTRODUCTION

Gamified learning is an educational approach that uses game design elements in non-game learning activities to engage and motivate students. With the rapid advancement of AI technology, the emergence of promising applications makes game development via textual interactions accessible to more developers, with the additional promotion of natural language interfaces for AI-generated games.Based on the technological breakthroughs of AI-driven game design, we designed a scenario-based simulation game about the application of medical law programmed by AI on the Rosebud platform to explore the effectiveness of integrating educational playfulness and gamified learning.The main goals are: (1) To assess the feasibility of designing games fully driven by AI using natural language. (2) To explore the potential integration of games and pedagogy. (3) To assess students‘ learning achievement and motivation using a survey. The basic novel contribution of this paper is to explore the possibilities and effectiveness of applying natural language-driven AI game editing to educational curricula.

2 LITERATURE REVIEW

2.1 Theoretical Foundation of Gamified Learning

Gamification methods introduce certain game design aspects such as points, levels and challenge problems in order to attract students' learning activities and to stimulate them as an internal motivator. The literature reveals that gamified learning significantly improves the efficiency and engagement of learning.Nevertheless, there are also issues—such as primarily related to difficulties in adapting learning mechanisms to specific educational intentions. [2]Despite these reservations, the fact is that gamification methods based in the extensive use of technology nowadays (such as AI and IT) are a deep perspective for educative implementation in the future since they are also much more applied.

2.2 Current Development of AI Technology and Application of Natural Language-Driven AI Programming

For several decades, the academic and industrial sectors have exhibited a persistent focus on the application of natural language in programming. Due to the rapid advancement of AI, an innovative approach has emerged. The past few years' development in AI technology, especially in sophisticated natural language processing (NLP), have now enabled the notion of creating a natural language driven programming code generation solution.As seen in Rosebud AI, Natural Language Processing has been a good solution to reduce the development difficulty in building games and motivate the nonexperts to prototype games quickly, thus mass-market programming.[3]

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2.3 Potential and prospects of NLP enabled Artificial intelligence in Future of education and Gamification based learning

Natural language-based AI programming is relatively accessible due to the nature, and thus may be able to become one of the mass innovative learning technologies. The basic attraction is its obvious ability of lowering the traditional programming barrier for general educators, parents, students with limited programming experience and technical knowledge easily design personalized and multi-variation interactive learning activities, game-like activities, and educational digital resources.[4]Even though still very much at an immature state, this accessibility and flexibility seem to suggest a wide potential that can significantly help gamified learning development allowing educational stakeholders to rapidly, and with minimum effort, design engaging and educational material.[5]

3 METHODOLOGY

3.1 Scenario Design Based on Natural Language Interaction

We used the Rosebud AI platform, programming through natural language-driven AI, to design interactive medical legal scenario simulation games. Through natural language instruction inputs, researchers could guide game content generation. For example, Create an examination room scene that includes an informed consent segment.

Similar to Rosebud AI, AI programming platforms can be viewed as multimodal generative models.[6] The technical implementation process is divided into five modules:

1.Natural Language Input.

2.Intent Recognition and Task Decomposition.

3.Code Generation.

4.Real-time Preview and Editing.

5.Automatic Deployment and Hosting

3.2 Game design

3.2.1 Background and Significance of Game Design. In this paper, the purpose is to use an interactive scenario simulation game to guide medical students to study medical laws and regulations; therefore, as medical law teacher, I found that the design of the interactive scenario simulation game can better improve teaching effect. The designed game has been published on the Rosebud AI platform. Though the designed interactive scenario simulation game also has some inevitable disadvantages, it can bring unique and irreplaceable good effects.It has low threshold to entry, is accessible, and is efficient, which makes it particularly appropriate for more general adoption. Additionally, these properties all happen to fit particularly well with education's ultimate values of diversity, equity, and sustainability.

In fact, few teachers have the programming skills to create customized educational software currently, and they have to choose commercial or traditional education software. Thus, it is challenging to customize teaching content and teaching procedures in a flexible way to satisfy learners' needs. [7]These restrictions make gamified learning hard to be implemented comprehensively.Luckily, through the development in the AI technology, teachers who have no background in programming can independently develop educational software, which can boost personalized, entertained, active and game-based learning.

Thus far, although the AI-based programming still just a supplementary coding approach, resulting in more simple, operational, and not necessarily beautiful game products, it makes it possible for the teachers to implement simple, operational games highly correlated with curriculum requirements.[8]In this research, we entirely accepted the game design by AI following natural language specifications and mainly validate two hypotheses: Firstly, teachers without programming skills can readily employ AI and natural language to produce real-world educational software or games to provide a broader range of gamified learning and investigate potential learning possibilities in the future.Second, it could deliver real advantages to students and get their appreciation, which could greatly enhance students' learning motivation and acquisition of the knowledge of the subject.[9]

3.2.2 Game Design Principles and Concepts. The above four elements mainly consist of game architecture, scoring rules, scenarios, and educational content.

Game structure consists of interactive role-playing scenarios based on dialogue and conversation

The medical environment is assumed to have three situations, which are regarded as the common scene of the medical site: emergency room, examining room, operation preparation room

Participants play the role of doctors, making judgments in clinical scenarios

Basic educational content clearly focuses on knowledge areas required for China's National Medical Licensing Examination.

Implementing the Game Using Natural Language-Driven AI Programming

We have actually implemented the game using AI via the Rosebud platform. Herein we present exact specifications and detailed demonstrative description on how this natural language-enabled AI methodology was instantiated in practice.

(1) Game Structure:

Three different scenarios: emergency room, examination room, surgical preparation room

Each scenario contains three independent questions

Every question possesses three possible choices: one correct and two wrong

Participants accumulate points by selecting correct answers

Correct selections earn 10 points and advance to the next question; incorrect selections provide legal compliance feedback before advancing to the next item.

Each question allows one answering opportunity; after an incorrect selection, legal tips are provided, students read them and automatically proceed to the next question with no opportunity to answer again.

Scene switching functionality allows players to freely select and switch between scenarios.

(2) Scoring System:

Initial score is zero

Correct answer: +10 points

Incorrect answer: 0 points, displaying the prompt: "No points earned."

Total score displayed upon completion

(3) Scenario and Dialogue Design: Scenario 1 (Emergency Room)

Background: Patient unconscious following a critical accident, needs care and treatment now!

Typical example questions and options are dealing with informed consent in an emergency, respecting a patient's autonomy, and disclosure of medical records.

Scenario 2 (Examination Room)

Background: Chronic disease patient consultation, involving privacy and informed consent.

Questions involve confidentiality responsibilities, clarity of side effects and risks, and procedures for using experimental drugs.

Scenario 3 (Surgical Preparation Room)

Background: Surgical preparation requiring clear informed consent and sound medical responsibility.

Includes risk disclosure details, decisionmaking process in case of unforeseen intraoperative findings, and post-operative patient complaint reactions.

(4) Visual Interface and Interactive Elements:

Backgrounds designed as static two-dimensional images reflecting realistic hospital environments:

Emergency room: urgent hospital setting, flashing medical lights, patient on stretcher.

Examination room: warm atmosphere, tables and chairs, seated patient.

Operating room: surgical suite, sterile work areas, lights, patient on table.

Interactive characters: doctor character in white coat interacting with patients.

Readout with the current score; selectable by buttons

Incorrect selections trigger pop-up windows with explanations

We implemented this game in the Rosebud AI platform, which can access the computer or cell phone. Answer selection can be made by clicking buttons.

(5) Additional Requirements:

Frontend for introduction screen: Hello! Welcome to Healthcare Law Education Game. You are a doctor. You will have the chance to learn laws about healthcare via situations.

Score summary screen with total score and positive message : The score of this run : X (of 90 points)Well done!

This design provides a meaningful pathway for effectively integrating natural language-driven interactive simulations in educational contexts, advancing the teaching frontiers of current medical legal education.

4 EXPERIMENTAL DESIGN

4.1 Experimental and Control Group Design

Experimental Group (Gamified Teaching Group):

Number: 101 students

Teaching Method: Interactive scenario simulation games for medical law designed using the Rosebud AI platform

Practice Session: 1- Session / Week (3 weeks) Time: 10 minutes / Session Practice Period

Control Group (Traditional Teaching Group):

Number: 100 students

Teaching Strategy: Lecture via Powerpoint & Traditional Lecture.

Class Meeting: 40 min/week, over 3 weeks.

Questionnaire Validity and Reliability Pre-test Explanation:

4.2 Validity Testing

Asking 5 medical law experts and practicing on-site teachers to verify the measurement items of the questionnaire whether measured the teaching effect of students or student attitudes.

Selected 10 students for small-scale interviews and preliminary investigations to revise the test

4.3 Reliability Testing

Used Cronbach's α coefficient to test questionnaire internal consistency, with α values above 0.9.

4.4 Assessment Methods

We employed quantitative questionnaires and qualitative feedback to study and measure the effectiveness of natural language-driven AI interactive gamified learning methods compared to traditional teaching.

5 EXPERIMENTAL MEASUREMENTS AND QUETIONNAIRE

After completing the game design, 207 medical students were recruited to experience the proposed healthcare regulations scenario simulation game. After interacting with the game, each participant completed a questionnaire evaluating their attitudes toward the educational game and the overall gamified learning approach. The specific content of the questionnaire was as follows:

I. Basic Information:

Your gender: □ Male □ Female

Your frequency of playing electronic games: □ Often □ Occasionally □ Almost never

Have you received systematic medical legal course learning before? □ Yes □ No

II. Teaching Effect Evaluation Items (Likert five-point scale, from 1 completely disagree to 5 completely agree)

This teaching method can promote my active participation in problem interaction

This teaching method effectively improved my understanding and memory of medical legal knowledge

This teaching method increased the fun and appeal of medical legal knowledge

This teaching method enhanced my learning motivation and interest

This teaching method enhanced my confidence in independently solving practical medical legal problems

I am willing to continue using this method to learn other medical legal knowledge in the future

This teaching method helped me master the key points of medical legal knowledge more solidly

III. Gamified Teaching Detail Evaluation Items (Only gamified learning group to complete) (Likert five-point scale, from 1 completely disagree to 5 completely agree)

Table 1: Comparison of Baseline Characteristics Between Two Groups of Participants

Variable	Category	Gamification Group (N=101)	Traditional Group (N=100)	Statistic (χ^2)	p-value
Gender	Male	58 (57.43%)	56 (56%)	0.038	0.845
	Female	43 (42.57%)	44 (44%)		
Prior experience with medical law courses	Yes	80 (79.21%)	69 (69%)	2.71	0.100
	No	21 (20.79%)	31 (31%)		
Gaming frequency	Often	32 (31.68%)	52 (52%)	10.09	0.006
	Sometimes	41 (40.59%)	34 (34%)		
	Never	28 (27.72%)	14 (14%)		

The game's settings are realistic and clear, making it easy to understand medical legal knowledge

The game's point system effectively motivates me to answer seriously and think deeply

The learning frequency of once a week, 10 minutes each time is appropriate and effective for me

The game's interactive information presentation helps me actually remember medical legal knowledge

The feedback provided for incorrect options (explanations of legal rules) is very helpful to me

I would prefer this learning method more if the game interface design were more exquisite and detailed

Compared to traditional classrooms, gamified interaction more easily guides me to participate actively

The gamified teaching form encourages me to actively explore more medical law-related knowledge

The game method allows me to experience more realistic application scenarios for medical legal cases

My interest in learning would be higher if more complex medical legal cases were added

IV. Open-ended Feedback and In-depth Interview Willingness (Not required)

Please indicate what you were most satisfied with regarding the learning method you experienced (gamified or traditional)

Please specifically indicate what needs improvement in the learning method you experienced (gamified or traditional)

6 DATA ANALYSIS AND DISCUSSION

6.1 Data Overview and Descriptive Statistics

This study assessed student attitudes towards healthcare law scenario simulation games and gamified educational methods. From an initial 207 questionnaires collected, 6 were excluded as invalid, resulting in 201 valid responses for analysis (gamified group: n=101; traditional group: n=100). The demographic characteristics of the participants are summarized in Table 1.

6.2 Reliability Analysis

To assess the internal consistency of the questionnaire's multi-item scales, Cronbach's α reliability analysis was performed using SPSS software. This analysis specifically targeted two key sections: "Perceived Effectiveness of Teaching Methods" (Part II) and "Gamified Teaching Method Detail Measurement" (Part III). The reliability coefficients are presented in Table 2.

Reliability analysis results show that the measurement tools adopted in this article possess adequate internal consistency with reliable measurement results, which would enable follow-up data analyses.

6.3 Descriptive Statistical Analysis

We calculated the means and standard deviations of the main variables (especially the items in Parts II and III) to understand participants' overall evaluation levels of each content. Table 3 shows the descriptive statistical results of the core items in Part II between the two groups. The means and standard deviations of all attitude constructs were determined. The results are shown in Table 3.

6.4 Difference Analysis

To test the core hypothesis of this study—whether the gamified learning method is superior to traditional teaching methods in enhancing students' perception of learning effectiveness—we used independent sample t-tests to compare differences between the gamified learning group and traditional teaching group in each item and overall score of Part II "Perceived Effectiveness of Teaching Methods".The results are shown in Table 4, Table 5 and Table 6.

We found that compared with the non-gamified group, learners who received gamified learning obtained notably improved scores on all factors, especially on the understanding and remembering of knowledge (t=5.34, p<0.01, d=0.75) and deep understanding of key content (t=5.91, p<0.01, d=0.83); thus, the learning effect of natural language-driven AI programming is very positive and the use of this technology to increase the efficiency of education in complex professional fields, such as medical law, has been validated.

6.5 Correlation Analysis

We computed the Pearson's correlation coefficient between the total score in the Part II "Perceived Teaching Method Effectiveness" and the total score in the Part III "Gamified Teaching Method Detail Measurement" to determine the relatedness among evaluation dimensions among the gamified learning group and find out which game elements are strongly correlated with the perceived learning effectiveness in general.

A significant positive correlation (r=0.93,p<0.01) was found between students' perception of teaching effectiveness and evaluation

Table 2: Reliability Analysis Results of the Questionnaire Scales.

Construct	Section	Applicable Group	Items	Cronbach's α	Reliability Level
Perceived Effectiveness of Teaching Methods	Section 2	All Participants	7	0.932	High
Evaluation of Gamified Design Details	Section 3	Gamified Group Only	10	0.921	High
Perceived Usefulness (PU)	Section 4	Gamified Group Only	4	0.927	High

Table 3: Descriptive Statistics for Perceived Effectiveness by Group.

Item	Group	N	Mean	Standard Deviation (SD)
1. Active Interaction	Game	101	3.50	1.21
	Trad.	100	2.64	1.32

Table 4: Comparison of Perceived Effectiveness Between Gamified and Traditional Groups.

Item/Construct	Test Used	Gamified Group M (SD)	Traditional Group M (SD)	t-value	p-value	Effect Size (Cohen's d)
1. Active Interaction	T-test	3.50 (1.21)	2.64 (1.32)	4.87	p < 0.01	0.69
2. Knowledge & Memory	T-test	3.39 (1.25)	2.40 (1.39)	5.34	p < 0.01	0.75
3. Interest & Fun	T-test	3.48 (1.26)	2.60 (1.11)	5.33	p < 0.01	0.75
4. Motivation Boost	T-test	3.44 (1.26)	2.52 (1.39)	4.96	p < 0.01	0.70
5. Problem-Solving Confidence	T-test	3.29 (1.30)	2.51 (1.39)	4.09	p < 0.01	0.58
6. Future Willingness	T-test	3.01 (0.99)	2.60 (1.21)	4.01	p < 0.01	0.61
7. Mastery of Core Concepts	T-test	3.33 (1.28)	2.28 (1.25)	5.91	p < 0.01	0.83
Overall Score	T-test	3.34 (1.31)	2.51 (1.28)	5.61	p < 0.01	0.79

Table 5: Gender Differences in Attitudes Toward Gamified Scenario Design.

Variable	Gender	N	Mean	Standard Deviation (SD)	t-value	p-value
Evaluation of Game Design Details	Male	58	3.48	1.46	-0.316	0.752
	Female	43	3.58	1.44		

Table 6: Gender Differences in Attitudes Toward Gamified Learning.

Variable	Gender	N	Mean	Standard Deviation (SD)	t-value	p-value
Evaluation of Game Design Details	Male	58	3.48	1.46	-0.316	0.752
	Female	43	3.58	1.44		
Perceived Effectiveness (Gamified Group)	Male	58	3.73	1.42	0.764	0.447
	Female	43	3.51	1.32		

Table 7: Correlation Between Perceived Effectiveness and Evaluation of Game Design in the Gamified Group.

ariables Compared	Pearson Correlation (r)	p-value
Overall Effectiveness & Game Design Details Score	0.93	< 0.01

of game design details which meant that higher evaluation of game design details led to better perception of overall educational worth of the gamification. The results are shown in table 7.

6.6 Discussion

The conclusion derived from the analysis of this research shows that students have a relatively positive view of healthcare law scenario simulation games and the gamified teaching method behind. This is consistent with the first research hypothesis: gamified teaching modes driven by natural language are conducive to improving the class activity of students and their motivation for learning.

First, two essential constructs of questioner survey of ”Teaching Method Effectiveness perceiving” and ”evaluation of gamified teaching detail ”both have high reliability, this suggests that measuring tools have a good internal consistency and can explain well. Mean scores for the two constructs also shows description statistics of 3.34 and 3.52 respectively, showing that the students were in general satisfied with this teaching model but with a moderate/positive evaluation.

Second, the correlation analysis (Table 7) indicates positive correlations between “Perceived Teaching Method Effectiveness” and “Game Design Details Evaluation” ($r=0.93$, $p<0.01$) which means the students evaluation on the teaching method as a whole has similar level of consistency with the perceived satisfaction of individual game design details.According to this result, carefully crafted situations for interaction and feedback might be central to optimizing the experience and outcomes of learning.

Moreover, no noticeable effect was seen from the gender variables, as far as it seems the approaches for teaching are highly similar between male and female students ($p>0.05$), which confirm the robustness and lack of bias on this approach at the gender scale.Despite that, from an educational aspect, although no significance test was conducted, preliminary findings reveal that Master’s and PhD’s graduate-level students have slightly better evaluation scores on gamified learning than undergraduates’ ones, which could be due to the fact that higher-year students have relatively better understanding of the realistic meaning and practical significance of simulation-based learning when they are confronted with various complicated legal circumstances.

In summary, this paper has proven the feasibility and teaching effect of using natural language based AI programming for building medical legal education scenes, and all aspects of the students’ comprehension, involvement, and interest in teaching contents have received great improvement, demonstrating that this approach has sufficient applicability and dissemination value in college education.Future study may take into consideration of the enhancing roles of the game difficulty adaptation mechanism, case complexity ranking mechanism, and multiple-round interactive feedback mechanism in further improving learning effect.

7 CONCLUSION

The paper has discussed the viability and efficacy of the AI programming game used in the gamified learning scenarios as a widely adoptable approach. The research revealed that this novel learning paradigm provides a personalized and effective way which has demonstrated a potential for wide application. Additionally, the widespread applicability and the rapid advancement of AI programming promote educational equity by means of reducing the barriers of designing gamified learning scenarios.

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